

Project 8 — Transcriptomic Analysis of Chemotherapy Response in Breast Cancer

A GEO-based computational biology study using GSE162187

Abstract

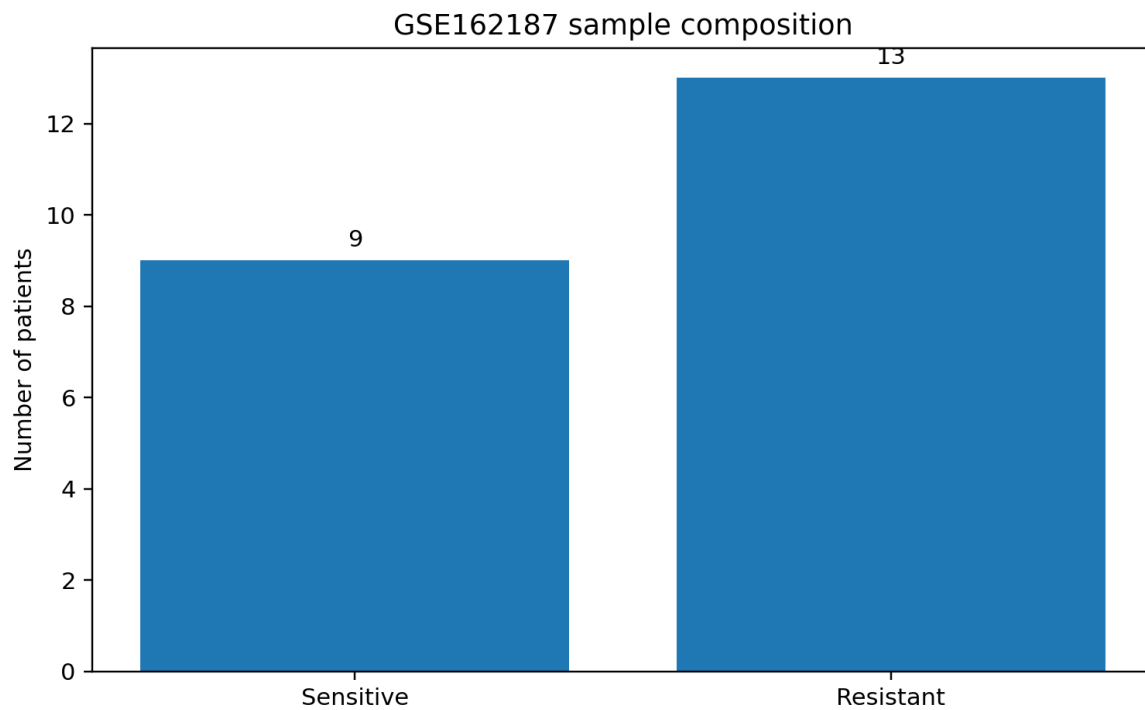
This project examines publicly available RNA-seq data from breast cancer patients who subsequently received neoadjuvant chemotherapy. The study compares treatment-sensitive and treatment-resistant patients and uses the published computational results as a reproducibility benchmark. The dataset contains 22 patient profiles: 9 classified as sensitive and 13 as resistant. The original analysis reported 1,985 differentially expressed genes (DEGs), followed by functional analysis and survival-oriented validation. This portfolio project focuses on understanding the computational workflow, interpreting the reported results, and connecting transcriptomic differences with chemotherapy response and potential biomarkers.

Research Question

Which transcriptomic differences are associated with chemotherapy sensitivity or resistance in breast cancer, and which candidate genes show evidence of prognostic relevance?

Dataset

Resource	GSE162187 / PRJNA680808
Organism	Homo sapiens
Experiment	RNA-seq gene-expression profiling
Samples	22 breast cancer patient profiles
Sensitive	9
Resistant	13
Genome reference	GRCh38
Repository	NCBI Gene Expression Omnibus (GEO)

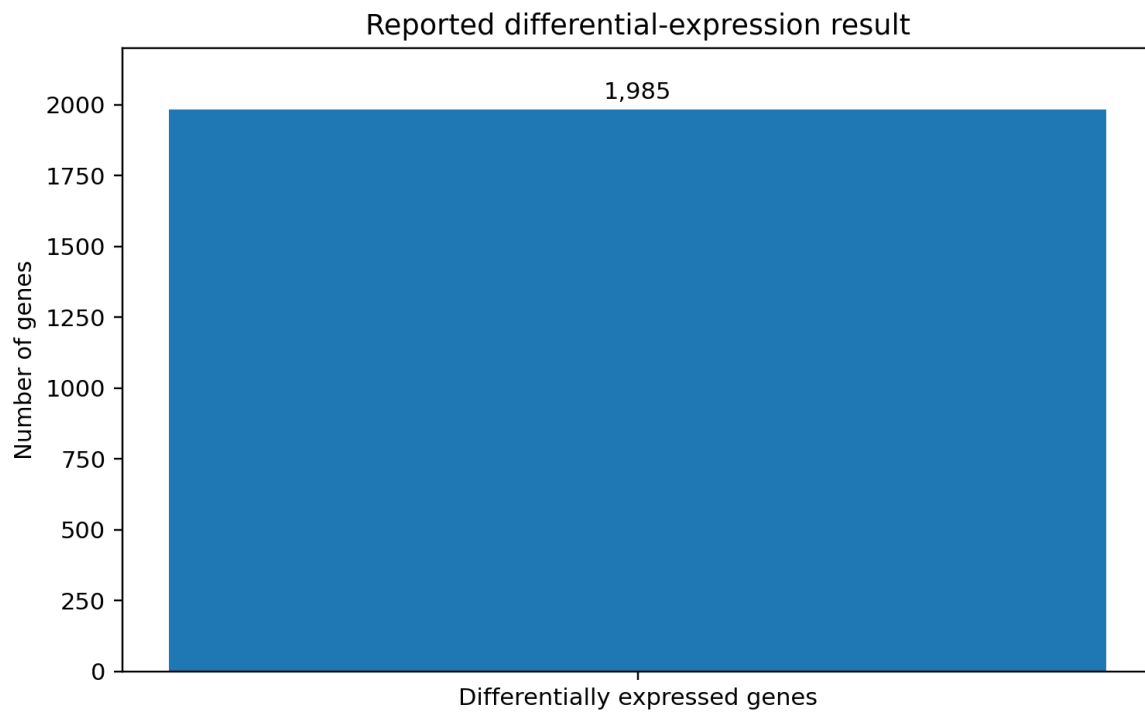


Computational Workflow

RNA-seq reads were processed by the original study using adapter filtering, Kallisto-based mapping/quantification against GRCh38, and DESeq2 for differential expression. The sensitive group was used as the reference when evaluating response-associated expression changes. Multiple-testing correction was performed using the Benjamini–Hochberg procedure. Functional interpretation included Gene Ontology, KEGG, GSEA and protein-association analyses.

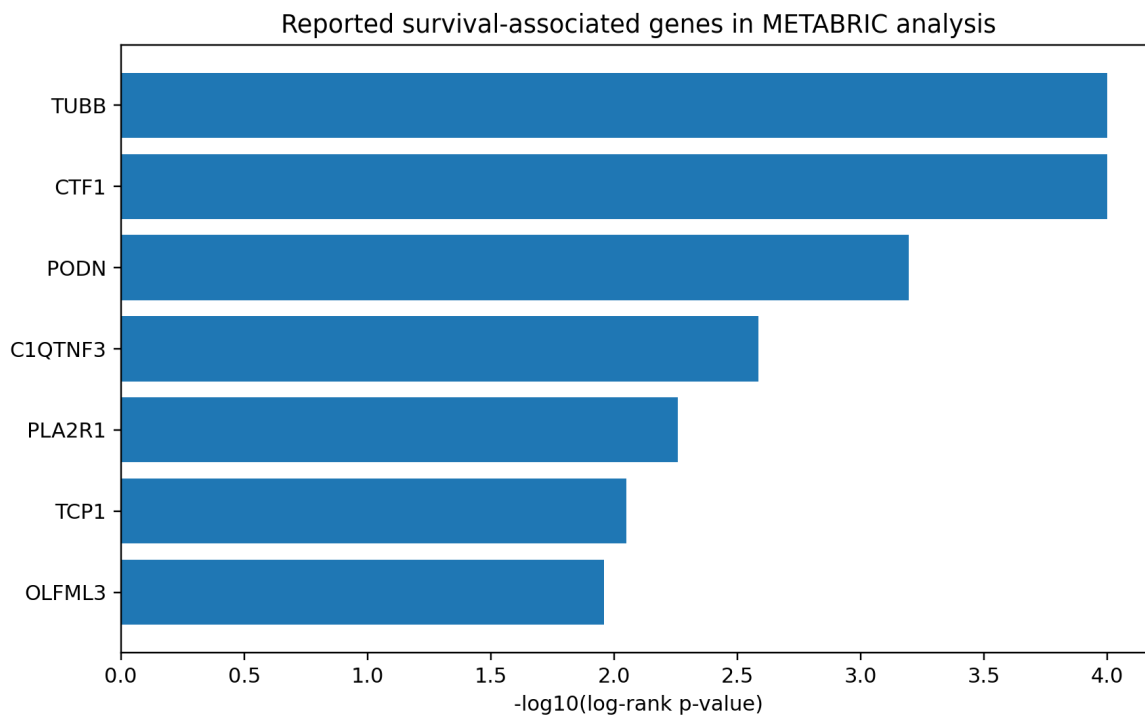
Differential-Expression Result

The original analysis reported 1,985 differentially expressed genes between chemotherapy-sensitive and chemotherapy-resistant patients. Reported enriched themes included metabolic pathways, pathways in cancer, cytokine–cytokine receptor interactions and neuroactive ligand–receptor interactions.



Candidate Biomarker and Survival Analysis

From the differential-expression results, 73 genes were carried forward for survival analysis using external breast cancer datasets. In the METABRIC analysis, seven genes showed statistically significant log-rank results: TCP1, TUBB, C1QTNF3, CTF1, OLFML3, PLA2R1 and PODN. The reported log-rank p-values ranged from less than 0.0001 to 0.011. A separate TCGA analysis identified KRT15, HLA-A and TCP1 as significant in that dataset.



Cross-Dataset Validation

The original study compared its 73-gene set with GSE163882. Eleven genes were reported in common between the two datasets: C1QTNF3, PODN, TUBB, MRGPRF, PLA2R1, KRT15, HOXD8, DCN, SOD3, SLIT3 and PRAME. This type of cross-dataset comparison is important because a signal observed in one cohort is more convincing when it is also observed independently.

Biological Interpretation

The results support the idea that chemotherapy response is associated with broad transcriptomic differences rather than a single gene acting alone. The reported pathway findings point toward changes involving metabolism, cancer-related signalling and cell–cell communication. The survival analyses further illustrate how transcriptomic candidates can be taken from an initial discovery stage toward biomarker evaluation.

Important Methodological Note

For this portfolio report, I have deliberately separated independently verified dataset facts from the original study's reported analytical results. The GEO record confirms the 22-sample study, the sensitive/resistant design and the availability of processed expression data. Because the raw supplementary matrix could not be executed in this environment as a local reproducible DESeq2 run, the numerical DEG and survival results above are presented as a reproducibility benchmark from the published study, not as newly generated statistics. This avoids presenting another researcher's results as if they were independently reproduced.

What I Learned

- How GEO can be used as a source of public transcriptomic datasets.
- Why sample grouping and reference-group selection matter in differential expression.
- Why raw counts require appropriate count-based statistical methods such as DESeq2.
- How multiple-testing correction helps control false discoveries.
- How differential-expression results can be connected to pathway and survival analyses.
- Why independent datasets are valuable for validating candidate biomarkers.

Limitations

- The cohort is small (22 patients), so statistical power is limited.
- Treatment regimens and patient biology may introduce heterogeneity.
- Association does not establish that a candidate gene causes drug resistance.
- Published DEG results should not be treated as independent validation.
- A stronger follow-up would reproduce the count-level analysis locally and then validate candidates in an independent cohort.

Conclusion

This project demonstrates a complete computational-biology reasoning chain: public clinical transcriptomics → differential expression → functional interpretation → candidate prioritisation → external validation. The main lesson is that bioinformatics is not simply about generating a list of significant genes; the value comes from asking a biological question, controlling statistical error, interpreting the results carefully and testing whether signals remain meaningful in independent data.

Sources

- NCBI GEO: GSE162187 — Transcriptomic analysis of breast cancer patients sensitive and resistant to chemotherapy.
- Barrón-Gallardo CA et al. Transcriptomic Analysis of Breast Cancer Patients Sensitive and Resistant to Chemotherapy: Looking for Overall Survival and Drug Resistance Biomarkers. *Technology in Cancer Research & Treatment*. 2022;21.
- NCBI BioProject: PRJNA680808.
- NCBI GEO sample metadata and processed expression-data records.